# RUDRESH ASAGODU

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## EDUCATION

**Msc Physics + B.E. Electronics and Instrumentation** August 2023 - Current
Birla Institute of Technology and Science Pilani Telangana, India

## EXPERIENCE

**Robotics and Intelligent Systems** PNT robotics, Mumbai, India
*Summer Internship* May 2025- August 2025

- Implemented a real-time hand-tracking and gesture recognition using OpenCV for robotic hand control and sign-language detection, enabling recognition of 10 alphabetic hand gestures.
- Constructed a robotic hand mimic system integrating hand detection, gripper mechanisms, and communication protocols including UART, SPI, and I2C.
- Designed an autonomous guided vehicle (AGV) control system for line following and junction detection using LSA08 line sensors and sensor-based feedback.

## PROJECTS

**Monte Carlo Simulation and Computational Methods**
*Study Oriented Project under Professor Sutapa Roy* Faculty website

**ML and Optimization**
*Design Oriented Project under Professor Pratyush Chakraborty* Faculty website

- Engineered a ML based approach for optimizing data-centre energy consumption using energy storage systems.
- Investigated carbon-reduction strategies for LLM workloads and evaluated the potential of SLMs as lower-emission alternatives for tasks such as Image Captioning, Visual Question Answering, Dialogue Summarization, and Text-to-SQL conversion.

*Reference: "Emissions and Performance Trade-off Between Small and Large Language Models" (FLLM 2025).*

## CERTIFICATIONS

**Machine Learning with Python** Link

- Studied fundamental machine learning concepts and algorithms using Python.
- Worked with supervised and unsupervised learning techniques.
- Applied machine learning methods to real-world datasets through coursework and projects.

**Introduction to Deep learning and Neural Networks** Link

- Studied fundamentals of deep learning, neural networks, and model training.
- Implemented regression and classification models using the Keras library.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, JAVA, LaTeX, MATLAB

**Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn

## RELEVANT COURSES

- **Probability & Statistics:** Probability and Statistics, Applied Stochastic Processes
- **Computational & Programming:** Computational Methods in Physics, Object Oriented Programming
- **Finance:** Derivatives and Risk Management
- **Physics:** Statistical Mechanics